

Despacho Laboral — Complete System & Features Document

System: Web application for managing labor law cases at a law firm in Tijuana, Mexico **Product name:** Conciliación Laboral Tijuana — Despacho Laboral **Stack:** Django 5.x · Tailwind CSS · HTMX · PostgreSQL/SQLite · Celery + Redis · Selenium **Last updated:** August 2026

1. What the System Does

Despacho Laboral is a **case-management (CRM + legal workflow) platform** for a labor law practice. It manages the complete life cycle of a labor claim — from the first client contact, through the mandatory pre-trial conciliation stage with the *Centro de Conciliación Laboral (CCL)* of Baja California, up to settlement (convenio) or filing a formal lawsuit (demanda laboral).

Beyond case tracking, the platform automates the most repetitive and error-prone parts of the practice:

- **Legal benefit calculations** (aguinaldo, vacation premium, seniority premium, constitutional severance, etc.) computed automatically from Mexican labor law (LFT).
- **Legal document generation** from reusable HTML templates (machotes) with automatic data injection.
- **Automated submission of conciliation requests** to the official BC government portal (`app.conciliacionbc.gob.mx`) — both server-side and via a companion Chrome extension.
- **WhatsApp notifications** to clients (manual, automatic per state change, and reminders).
- **A full finance module** (payments, expenses, payroll, commissions, cash registers, partner loans, profit distribution) organized per office and per work week.

2. Users, Roles & Permissions

2.1 Roles (5)

Role	Key	Scope
Superadmin	superadmin	Full access: Django admin, all dashboards, permission matrix, user management
Administrativo	admin	Sees all cases, productivity reports, totals, Excel exports, Django admin access
Asesor	asesor	Only sees and edits his own cases; personal dashboard with stats
Abogada	abogada	Dedicated dashboard for the firm's attorney (case overview & document focus)
Finanzas	finanzas	Financial dashboard and finance module access (counts as admin for permissions)

2.2 Profile capabilities (`UserProfile`)

Each user profile additionally has:

- ``puede_generar_documentos`` — flag granting access to the legal document generators (demanda, machotes, legal documents). Off by default.

- ``api_token`` — personal API token used by the Chrome extension to authenticate against the app's extension API. Auto-generated on profile creation and can be regenerated (invalidating the old one) from the **Extensión de Chrome** config page.

2.3 Permission audit (``PermisoAuditLog``)

Every time a superadmin/admin changes a user's role, document permissions, or other privileges, an **audit entry** is recorded: who was modified, who modified them, the action (`cambio_rol`, `cambio_docs`, `mixto`), and the detail (e.g. `Rol: asesor → admin`).

2.4 Superadmin module

- **Superadmin dashboard** — overview panel for platform administration.
- **Permission matrix** (`matriz_permisos`) — grid to manage every user's role and document-generation permission, with **Excel export**.
- **Load demo data** — one-click seeding of demo users/cases for testing.

2.5 Auth & security

- Login / logout, full **password reset** flow via email (SMTP configurable; console backend by default in dev).
- Redirect to the correct dashboard automatically per role after login.
- Production hardening: HTTPS redirect, secure cookies, Railway domain auto-detection, env-driven `DEBUG / SECRET_KEY / ALLOWED_HOSTS`.

3. Case Management (Módulo Expedientes)

3.1 Clients (``Cliente``)

Rich client record built specifically for labor conciliation:

- **Identity:** name, **CURP** (unique), RFC, phone, WhatsApp, email, birth date, gender.
- **Home address:** street, number, ZIP, colonia — assembled into `direccion_completa`.
- **Employer data:** company name, economic activity, phone, legal name, full address and references.
- **Employer person type** (física / moral) — used by the conciliation portal form.
- **Employment data:** position, **monthly salary**, pay period (daily/weekly/biweekly/ monthly), weekly hours, work shift (diurna/nocturna/mixta), hire date, termination date.
- **Marketing/source:** how the client heard about the firm (Facebook, Google, recommendation, WhatsApp, TikTok, TV/radio, flyer...).
- **Office assignment** (Plaza Patria, Otay, CLT).
- **Free consultation tracking:** whether the free weekly consultation was offered, scheduled, and on which date.
- Indexed for fast lookups by CURP, name, and consultation date.

3.2 Case files (``Expediente``)

- **Automatic numbering** `AAAA-####` (year + sequential), unique and non-editable.
- Assigned to one **asesor** (legal advisor).
- **Amounts:** claimed amount (`monto_reclamado`) and settlement amount (`monto_convenio`).
- **Hearing scheduling** with date/time and recorded result.

- **Conciliation data:** dismissal type (justified/unjustified/voluntary/rescission/other), claimed benefits, **folio** and processing date.
- **Follow-up:** next action date, internal notes, priority (baja/media/alta).
- **Per-case WhatsApp automation toggle** (`notificar_whatsapp_auto`).

State machine with validated transitions

```
nuevo → solicitud → citatorio → audiencia → no_notificado ⊢
                                     ↳ reprogramacion ⊢
                                     ↳ convenio ————|
                                     ↳ sin_conciliacion ⊢ demanda → audiencia/convenio/cerrado
                                     ↳ cerrado (from any state)
```

The model **rejects illegal transitions** (e.g. jumping from `nuevo` to `audiencia`) with a clear message listing the allowed transitions. States have associated colors for UI badges.

3.3 Documents (`Documento`)

- Upload PDFs/images per case with type classification (INE, contract, evidence, screenshot, citatorio, PDF, other) and description; uploader and date recorded.
- Organized by year/month folders on disk.

3.4 Notes & Activity history

- **Notas** — free-form notes per case with author and timestamp.
- **Movimientos** — automatic audit trail of every action: creation, state change, update, document upload, note added, hearing result (who, when, what changed).

3.5 Notifications (bell)

In-app notifications for the user: transfers, avisos, system messages, reminders. Each notification can carry a link; mark single or **all as read**.

3.6 Transfer requests between advisors

When an asesor can't attend a hearing (schedule conflict, etc.):

- He requests a **transfer** of the case, optionally suggesting a destination asesor, with a mandatory reason.
- Administration reviews the request, optionally **reassigns** to another advisor, and approves or rejects with a comment.
- Statuses: pending → approved / rejected / cancelled. Notifications keep everyone informed.

3.7 Avisos (mandatory admin notices)

- Admin creates notices/pending items with title, content, priority, and **optional expiry date** (auto-hides after expiry).
- Shown on every advisor/admin dashboard; users can mark them as **read/understood** (tracked per user via M2M).

3.8 Global search & filtering

- **Global search** across cases, clients, folios, companies.
- Advanced filters on the case list; asesores only see their own cases.

3.9 Calendar

- **Hearing calendar** (`calendario`) aggregating all scheduled hearings.

3.10 Reports & exports

- **Excel export** of cases.
- **Admin reports** page (`reportes_admin`) — productivity, totals per advisor.
- **PDF per case** (`pdf_expediente`) generated with WeasyPrint.

4. Dashboards (per role)

Dashboard	What it shows
Asesor	Personal KPIs, his cases by state, upcoming actions/hearings, pending tasks, avisos
Abogada	Attorney-oriented overview of the practice's cases and documents
Admin	All cases, productivity reports, total amounts, avisos, export actions
Finanzas	Financial KPIs (see Finance module)
Superadmin	Platform administration panel + permission matrix

5. Legal Calculation Engine (Cálculo Laboral)

5.1 Configurable legal rules (`LegalConfig`)

All legal parameters are stored in the database and editable from the Django admin — **no code changes needed when the law changes**:

- **UMA** daily value, general minimum wage, **northern border zone (ZLF) minimum wage**.
- **Aguinaldo** days (min. legal 15).
- **Vacation premium %** (min. 25%).
- **Seniority premium**: days per year (12) + salary cap type (2× UMA / 2× SM / 2× ZLF SM) and multiplier.
- **Constitutional severance days** (3 months = 90).
- Only one config can be **active** at a time; saving one deactivates the others.

5.2 Vacation table (LFT 2023 reform)

The system implements the current Mexican vacation table (year 1 → 12 days, +2 per year up to 20, then +2 every 5 years), extrapolating beyond the table when needed.

5.3 Concepts calculated

Concept	Article	Type
Aguinaldo proporcional	Art. 87 LFT	automatic
Vacaciones proporcionales	Art. 76 LFT	automatic
Prima vacacional	Art. 80 LFT	automatic
Prima de antigüedad (with cap flag)	Art. 162 LFT	automatic
Indemnización constitucional (90 days)	Art. 50 LFT	automatic
Indemnización 20 días por año	Art. 50-II LFT	automatic
Vacaciones vencidas (manual days)	Art. 76 LFT	semi-auto
Horas extras (manual hours)	Art. 66-68 LFT	semi-auto
Salarios devengados (manual amount)	Art. 48 LFT	manual
Días festivos (manual days)	Art. 75 LFT	semi-auto

5.4 Per-case calculation (`CalculoLaboral`)

- One calculation per expediente, storing a **snapshot of inputs** for history.
- **Checkboxes** let the advisor choose which concepts to include (defaults sensible).
- **Manual override** for vacation days actually owed (when some years were already paid or enjoyed) — shown with an **override applied** flag.
- Shows daily wage, days worked, years of service, per-concept breakdown with legal references, applied-cap warnings, and **total**.
- **Auto-recalculation** when client/case data or legal rules change (`recalcular`).
- **Quick simulation** (`simulacion-rapida`) — instant estimate for a prospect without creating a case: enter salary + dates and get the full breakdown.

6. Legal Document Generation

6.1 Machotes (reusable HTML templates)

- Templates categorized as: **Demanda Laboral, Carta Finiquito, Convenio, Solicitud, Citatorio, Otro**; with jurisdiction (Federal / Estatal BC / Both), optional dismissal type, icon, active flag, and ordering.
- **Favorite** templates float to the top of the editor.
- Template body is HTML using **markers** like `{{ nombre }}`, `{{ empresa }}`, etc.

6.2 Marker injection engine (`marcadores.py`)

- ~40 markers with **real case data**: client, employer, case, dates (in Spanish, e.g. "1 de enero de 2024"), salary formatting, **and computed benefits** (aguinaldo, vacations, indemnity...) pulled automatically from the calculation engine.
- Missing data renders as visible placeholders (`[CURP]`, `[MONTO]`, ...) instead of blank or crashing — so the user knows exactly what to fill in.
- **Completeness checker**: lists which fields are complete/incomplete per section (Cliente, Empleo, Empresa, Expediente) with **direct edit links** and a completeness percentage.

6.3 Document workflow

1. Choose a machote from the catalog (or from the case).
2. **Prepare** — see which data is missing, click through to fix it.
3. **Edit/preview** the rendered document in the browser.
4. **Generate/download** as PDF.
5. Optionally **save the edited document as a new machote** (team template library grows organically).

6.4 Import machotes from Word

- `.docx` files (e.g. a folder of existing demandas) are converted to HTML templates via the `importar_machotes` command or web import, preserving markers where possible.
- Template library is managed from the Django admin or the machotes UI.

6.5 Demanda Laboral generator (`demanda\_generator.py`)

- Generates a **professional Mexican labor lawsuit document** (Word `.docx`) with all case data, integrated calculations, print/filing-ready formatting.
- A **step-by-step wizard** (`demanda_asistente`) guides through sections of the demanda; a **direct generator** produces the document in one click.
- Downloads as `.docx` for editing/signing.

7. Conciliation Automation (Portal de Baja California)

The system integrates with the official portal `app.conciliacionbc.gob.mx` in **two modes**.

7.1 Server-side automation (`conciliacion\_automation.py` + Celery)

- Fills the real multi-step portal flow: privacy notice → industry → conflict date & object → worker (solicitante) → employer (citado) → description → submit.
- Runs **headless** (server) or **visible/debug** mode.
- Executed asynchronously via **Celery worker** (or threading fallback without Redis) so HTTP requests never time out.
- Live progress page (`conciliacion_procesando`) showing **screenshots of the browser** and elapsed time; retry support.
- On success: captures the **folio**, saves the **acuse PDF** into the case, marks TareaConciliacion as completed with full logs.
- Management command `enviar_solicitud_conciliacion` for batch/CLI sending.

7.2 Acuse parser (`acuse\_parser.py`)

When an official accuse PDF is available (auto-downloaded or manually uploaded), the system **extracts its text with PyMuPDF** and auto-populates the case:

- Folio (e.g. `TIJ/26427/2026`), request date, applicant name, cited company, conflict date, object, conciliation unit.
- **Maps the conciliation object → dismissal type** (e.g. "Despido" → injustificado).
- **Preview screen** shows detected values vs. current values (new/differing flagged) and lets the advisor **confirm** before saving.

7.3 Chrome extension — "Conciliación BC Asistente"

For when automated navigation is detected or a CAPTCHA must be solved by a human, the companion **Chrome extension (Manifest V3)** fills the portal form **in the advisor's own browser**:

- The advisor clicks *Enviar a Conciliación → Desde mi navegador (Extensión de Chrome)*.
- The extension shows pending tasks (case number, client, company, CURP) in its popup.
- 🚀 **Llenar en el portal** opens the portal with the form **already filled** (privacy, industry, dates, worker, employer, description) via a content script — so the portal sees a human-driven session.
- The advisor reviews, solves CAPTCHAs live, and clicks *Enviar solicitud* himself.
- 🔍 **Ya envié, detectar acuse** — the extension detects the folio, downloads the acuse PDF, takes a screenshot, and **reports everything back to the app's API** (authenticated with the user's personal token), attaching the PDF to the case.
- Config screen (options) stores the app URL + token and verifies connectivity.
- The extension package can be **downloaded as a .zip** from the app (`/extension/descargar/`), ready to load unpacked into Chrome.

7.4 Manual fallbacks

- Upload the acuse PDF manually (`subir-conciliacion`), then preview/confirm the parsed data.
- Download a **pre-filled conciliation request PDF** to file manually at the center.
- Upload the acuse via the normal document flow.

8. WhatsApp Integration

Two delivery methods:

1. **Deep links (wa.me)** — free; opens WhatsApp with a pre-filled message.
2. **Twilio WhatsApp API** — server-sent messages (needs Twilio credentials); falls back to deep link automatically when not configured.

Features

- **Templates** for: hearing reminder, citatorio, convenio follow-up, general follow-up, document request, custom messages — with variables (`{cliente}` , `{fecha}` , `{asesor}` ...).
- **Automatic messages on state changes**: every case state has a configured message (new case, solicitud created, citatorio, hearing, not notified, rescheduled, convenio, no agreement, demanda, closed). Per-case toggle to enable/disable.
- **Message history** per case with status (pending/sent/failed), channel, and logs.
- Mexican phone normalization (adds country code 52, strips common formatting).
- Management command `enviar_whatsapp_automatico` to flush pending messages (with `--dry-run` and `--send-twilio` options).

9. Imports & Utilities

9.1 CLT import (citas from Excel)

`CLT.xlsx` (pre-scheduled conciliation appointments from the CCL) is imported as **expedientes with scheduled hearings**:

- Per-sheet import (e.g. monthly tabs), auto-creates **advisor users** found in the file if needed, and cleanup options for test data.
- CURP is optional at import time and captured later (so numbers flow in fast).

9.2 Company catalog (`Empresa`)

- Import from `Empresas y Domicilios.xlsx` — **idempotent**, names normalized (uppercase, no accents, compacted spaces) so re-importing a growing file never duplicates.
- Person type auto-detected; detailed address kept for **autocomplete in the client form** (`empresas/buscar/` AJAX endpoint).

9.3 CURP generator

Standalone tool to **generate a valid CURP** from personal data (name, birth date, gender, state) — saves manual lookups when onboarding clients.

9.4 User settings (Ajustes)

Per-user settings page: profile info, phone, WhatsApp automation preference, API token management, extension configuration.

10. Finance Module (Módulo Finanzas)

Organized by **office** (sucursal) and by **work week** for operational reporting.

10.1 Offices

Catalog of firm offices with address, phone, manager, active flag. Each office has its own income/expense/cash-register operations.

10.2 Work weeks

The finance system revolves around **weeks** (Mon–Sun): the current week is created automatically; every movement can be attributed to a week. Weeks can be opened/closed. Totals per week: **income** (cash + settlement payments), **expenses** (cash + expenses + payroll) and **balance** are computed on the fly.

10.3 Settlement payments (`SettlementPayment`)

Payments received from settlement agreements (convenios): date, client, case, amount, payment method (cash/transfer/check/credit/debit/deposit/other), office, notes, auditor.

10.4 Expenses (`Expense`)

Operational expenses per office with category (rent, utilities, internet, stationery, advertising, salaries, gas, maintenance, taxes, equipment, furniture, insurance, professional fees, other), provider, tax folio/invoice, and auditor.

10.5 Agreements & honorarios (`Agreement` / `Honorario`)

- **Convenios** with the client: amount, state (pending signature → signed → paid / partially paid / cancelled), responsible person, office, and automatic **total fees**.
- **Honorarios**: one or more fee entries per convenio with a **% tariff** (25/30/35/40/50%); amount auto-computed ($\text{convenio} \times \% \div 100$), expected/payment dates, paid/pending sums aggregated on the convenio.

10.6 Commissions (`Commission`)

Advisor commissions on closed convenios: % configurable, **amount auto-computed**, state (pending/paid/cancelled), payment date, office, auditor.

10.7 Employees & payroll (`Employee` / `Payroll`)

- Employee catalog with position (administrative, legal advisor, supervisor, accountant, receptionist, assistant, director...), pay period, salary, office, active flag.
- **Payroll payments** per period: period type (weekly/biweekly/monthly/extraordinary/ aguinaldo/vacation premium/bonus), period range, salary paid, deductions (ISR/IMSS), **total auto-computed** (`salario - descuentos`), office, auditor.

10.8 Cash register (`CashMovement`)

Daily cash movements per office and date: **income** (convenio payment, client payment, advance, refund, other) or **expense** (stationery, gas, rent, utilities, internet, phone, travel, commissions, fees, other). Category coherence with type is **validated** (e.g. an income can't have an expense category). Reference field links to case/invoice.

10.9 Partners, loans & profit distribution

- **Partners (`Partner`)**: participation % per partner for profit sharing.
- **Partner loans (`PartnerLoan`)**: formal record of loans between partners (origin → destination, amount, concept, status, pending balance), replacing informal IOUs.
- **Profit distribution (`ProfitDistribution` + `PartnerProfit`)**:
 - Auto-computes **net profit** = convenio amount – fees – commissions – withholdings – related expenses.
 - Distributes it among partners per their participation % (**individual share auto-calculated**), with states (draft → distributed → confirmed/cancelled).
 - Per-partner **accumulated summaries** (generated / paid / pending) auto-updated when distributions are confirmed.

10.10 Financial dashboard

- KPIs per office and overall; **monthly cash flow** chart fed by the `api_flujo_mensual` JSON endpoint; **Excel export** of the dashboard totals.
 - **Convenio reports** (`reporte_convenios`) with Excel export.
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11. Management Commands (CLI)

Command	Purpose
<code>crear_usuarios_prueba</code>	Create test users (1 superadmin, 4 admins, 15 asesores)
<code>seed_datos</code> / <code>seed_clientes_prueba</code>	Seed demo cases (10 labor scenarios / full clients with calculations)
<code>enviar_recordatorios</code>	Send upcoming-action/hearing reminders (<code>--days</code> , <code>--dry-run</code>)
<code>enviar_solicitud_conciliacion</code>	Submit conciliation requests to the BC portal from CLI (<code>--headless</code>)
<code>enviar_whatsapp_automtico</code>	Flush pending automatic WhatsApp messages (<code>--dry-run</code> , <code>--send-twilio</code>)
<code>importar_clt</code>	Import CLT.xlsx citas as expedientes with hearings
<code>importar_empresas</code>	Import the companies/addresses catalog (idempotent)
<code>importar_machotes</code>	Convert <code>.docx</code> templates into machotes (<code>--reload</code> , <code>--file</code>)
<code>migrate_sqlite_to_pg</code>	One-shot data migration from SQLite to PostgreSQL

12. Integrations Summary

Service	Use	Config
Selenium	Portal automation (headless/debug)	included
Celery + Redis	Background tasks (conciliation automation); threading fallback	<code>REDIS_URL</code>
Twilio	WhatsApp API sending (optional)	<code>TWILIO_*</code> env vars
SMTP	Password-reset emails (console backend in dev)	<code>EMAIL_*</code> env vars
WeasyPrint	PDF generation (expediente PDFs, documents)	included
python-docx	Word (.docx) demanda generation	included
PyMuPDF (fitz)	Acuse PDF text extraction	included
openpyxl	Excel imports (CLT, companies) and exports	included
Chrome extension API	Token-authenticated REST endpoints for the extension	personal <code>api_token</code>
Railway	Deployment; auto-detects domain, PostgreSQL and Redis env vars	env-driven

13. Architecture & Tech Notes

- **Backend:** Django 5.x, `es-mx` locale, `America/Mexico_City` timezone.
- **Database:** SQLite (dev) / PostgreSQL (prod) via `DATABASE_URL` ; Railway env auto-detection (`PGHOST` ...).
- **Frontend:** Tailwind CSS (CDN) + **HTMX** for dynamic interactions (no heavy JS framework); responsive design.
- **Static files:** Whitenoise (`CompressedManifestStaticFilesStorage`) for zero-Nginx production serving.
- **Background work:** Celery when Redis is present, **threading fallback** otherwise (graceful degradation for local dev).
- **Auditability:** every key entity (payments, expenses, payroll, cash, loans, distributions, commissions) records `registrado_por` / `creado_por` .

- **Deployment:** Dockerfile, Railway config, entrypoint (auto-starts Celery worker when Redis exists), Nginx + Gunicorn recipe for VPS (droplet), and client-facing deploy guide.
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14. Testing

The suite includes **48 tests** (expedientes, finanzas, automation) covering state transition validation, financial calculations, and the verification script `verify_demandas_calculos.py` (used by CI). Run with:

```
uv run python manage.py test
```

15. Quick Start (dev)

```
uv sync                                # install dependencies
uv run python manage.py migrate        # DB migrations
uv run python manage.py crear_usuarios_prueba
uv run python manage.py runserver
```

Test logins: `superadmin/Admin123!` · `admin1/Admin1!` · `asesor1/Asesor1!`
